Nuoxian Wang

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Education

Nanjing University (985) 2025/09 – 2027/06 (Expected)
Master, School of Intelligent Software and Engineering, Software Engineering

University of Science and Technology Beijing (211) 2021/09 – 2025/06
Bachelor, School of Computer and Communication Engineering, Internet of Things Engineering

Skills

Languages:
- Proficient: C/C++, Rust; Familiar: Python

Research Interests:
- Linux Kernel, RISC-V SBI, Database Kernel

Experience

Tencent Cloud, Big Data OLAP R&D Intern 2026/05 – Present
Maintained and developed Tencent Cloud TCHouse-D (based on Apache Doris), diagnosing and fixing FE/BE issues and contributing upstream. Both independently submitted PRs were merged into Apache Doris mainline.
- 63537 (BE / C++): Replaced the fixed configured interval used for Workload Group CPU and Scan IO rate calculations with the actual monotonic-clock interval, correcting metric inaccuracies caused by scheduling delays or runtime configuration changes; added division-by-zero protection and unit tests.
- 65659 (FE / Java): Reproduced and diagnosed a TOCTOU race in Nereids external-table partition pruning: the partition map was frozen during plan construction, while sorted partition ranges were re-read during pruning. A concurrent `ALTER TABLE ADD/DROP PARTITION` cache refresh could therefore mix snapshots, causing binary-search pruning to return a partition absent from the old map and trigger an NPE.

KernelSoft, OS R&D Intern 2024/01 – 2024/05
Participated in R&D of a self-developed intelligent cockpit real-time OS. Ported procfs-related features into the kernel, merged into the company's internal mainline.

Projects

SBI-Fuzz 2025/09 – Present
Individual research project: fully automated fuzzing tool for RISC-V SBI bootloaders. Auto-generates test cases from SBI spec interfaces and runs them on QEMU, with code coverage and seed mutation. Two RustSBI bugs found and confirmed by the community.

MiniOB 2024/09 – 2024/10
National College Student Computer Systems Capability Competition (OceanBase Database Competition) entry. Built a simplified database kernel as team lead, completing over half the tasks (update, B+Tree, expressions, functions). Perfect preliminary score; ranked 19th nationally, 3rd in Beijing. GitHub: https://github.com/bosswnx/miniob-2024

chaos 2024/01 – 2024/08
National College Student Computer Systems Capability Competition (OS Kernel Implementation) entry. Unix-like kernel in Rust based on Tsinghua rCore; multi-process, ext4, VisionFive 2. National Second Prize. GitHub: https://github.com/bosswnx/chaos

NJU Computer Systems Fundamentals Lab 2023/02 – 2023/05
Major lab in NJU–ICT(CAS) "One Student One Chip" program. Built NEMU (simplified QEMU) and Nanos-lite (paged time-sharing OS) covering RISC-V and OS fundamentals. Self-taught throughout; completed independently with deep understanding.

Awards

CSCC (OceanBase Database Competition)	Provincial 3rd Place	2024/12
CSCC (OS Kernel Implementation)	National 2nd Prize	2024/08
CCPC (Regional) Jinan	Bronze Medal	2023/12
ICPC (Regional) Nanjing	Bronze Medal	2023/11
MCM/ICM Mathematical Contest in Modeling	Honorable Mention	2023/02